AKSHAR VANDARA

Boston, MA | (857) 919-6234 | axr230102@gmail.com | LinkedIn | GitHub | Portfolio | Google Scholar

Software/Research Engineer • 3 peer-reviewed publications in PCG and constraint optimization

EDUCATION

Northeastern University — *M.S., Game Science and Design* **May '26**
GPA: 3.87 / 4.0 | Coursework: Game Data Science, Game AI *Boston, MA*

Gujarat Technological University — *B.E., Computer Engineering* **May '23**
GPA: 3.6 / 4.0 | Coursework: Algorithms, AI, ML *Ahmedabad, India*

SKILLS

Languages:	C++, C# (.NET), Python, JavaScript/TypeScript, Java, SQL, C
Backend / Cloud:	REST APIs, Google Sheets API, AWS S3, GCP, Firebase, Node.js, Flask, Django, React
Engine / Tools:	Unity (2D/3D), Unreal Engine 5, Git, Perforce, Docker, Xcode (iOS), ParaView/VTK
ML / AI:	PyTorch, TensorFlow, NumPy, Pandas, CUDA, Z3 SMT, PySAT, SAT/CSP modeling, OpenCV
CS Foundations:	Data Structures, Algorithms, Object-Oriented Design, State Machines, Multithreading

EXPERIENCE

Gameplay Programming Intern **Jul '26 – Present**
E-Line Media *Pheonix, AZ*

- Built a C# **content pipeline** (**Google Sheets API, AWS S3, Vimeo**) for an investor-pitch prototype; designers ship content without rebuilds.
- Designing a **data-driven state machine** for iOS game flow; designers reorder screens without code changes.
- Developed an **event-driven turn timeline** and a stack-based screen manager; and **save/load system** for player progression.
- Implemented a **pooled audio manager** with priority-based voice stealing; and scaled the iOS UI for iPhone and iPad.

Research Assistant **Feb '25 – Present**
Game Design Lab, Northeastern University *Boston, MA*

- Extended the **Sturgeon** level generator (**Python, PySAT**) to spacetime generation, running **100–500x faster** than Spacetime WFC; and designed an interactive editor for real-time level/path editing (*first-author, AAAI AIIDE 2025*)
- Added **4D spacetime blending** of learned game mechanics to the system; plus, **3D voxel visualization** via VTK/ParaView. (*first-author, FDG 2026 PCG Workshop*)
- Engineered a Python constraint solver (**Z3 SMT**) for generating **continuous spacetime** gameplay (*first-author, FDG 2026 PCG Workshop*).

Gameplay Systems & UI Programmer **May '25 – Aug '25**
Otisco Studios *Orlando, FL*

- Programmed **8+ core gameplay systems** in Unreal/C++ (including save/load and UI) across two titles.
- Created **5+ NPC and enemy AI archetypes** using Behavior Trees and custom EQS.
- **Doubled average framerate** by profiling and refactoring C++ gameplay code; resolved 30+ pre-launch bugs.

Unity Developer **Mar '23 – Jul '24**
Arcadon Games *Bangalore, India*

- Developed core gameplay mechanics in Unity/C# using state machines, improving input **responsiveness 30%.**
- Integrated Firebase to store and sync **1,000+** user profiles and inventory data, with sub-100ms response times.
- Wrote a match-simulation system that processes **30 matches/min** and cuts load times 40%.

PROJECTS

Spacetime PCG Research Suite [GitHub | Publication **] | *Python, PySAT, Pandas, Paraview***

- Open-source suite for 2D, 3D-blended, and physics-based level generation, with an interactive editor and VTK 3D views.
- Benchmarked generation speed and solvability across the three domains using Pandas, surfacing per-domain performance trade-offs.

Cricket Manager: League [Playstore **] | *Unity, C#, Firebase, Cloud Functions***

- Sports management game **live on Google Play** with ongoing updates.
- Ran alpha testing with **100+ users** across 3 major versions, reaching 85% satisfaction and **40% fewer bug reports**.

Red Circuit Rebellion [Steam] | *Unreal Engine 5, C++, Blueprints, Behavior Trees, EQS*

- Shipped on Steam with Story and Arcade modes, each with its own objective logic and enemy spawners.
- Implemented real-time environmental fracturing and a dynamic objectives manager in C++ and Blueprints.